

Cortex Speech

An offline-first desktop application for **Central Kurdish (Sorani)** speech transcription, transcript curation, and ML dataset export. The default transcription engine is the user's fine-tuned **OmniASR-7B Champion** running locally on the GPU; everything works without a network, and any cloud assistance is opt-in and off by default.

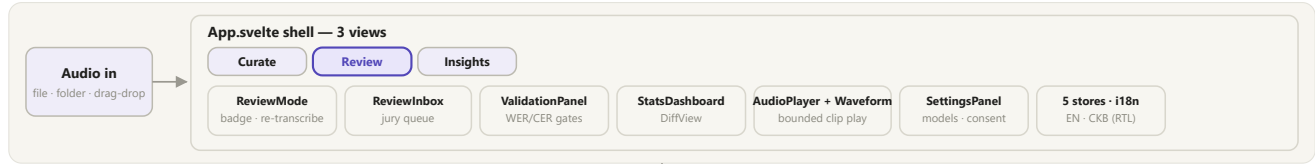
Architecture & technical overview · generated 2026-06-30 · repo github.com/HawzhinBlanca/cortex-speech · [main](#) @ [4787d81](#) · stack: Tauri v2 + Svelte 5 (runes) + Rust + SQLite/FTS5

Cortex Speech — End-to-End Architecture

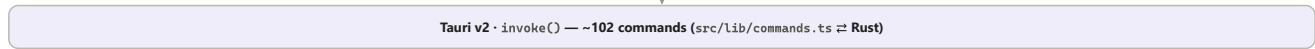
Offline-first Central Kurdish (Sorani) speech transcription & dataset curation · Tauri v2 + Svelte 5 + Rust · ~102 IPC commands · SQLite/FTS5

— Primary path · 7B Champion (default)
- - - - Cloud · opt-in, off by default

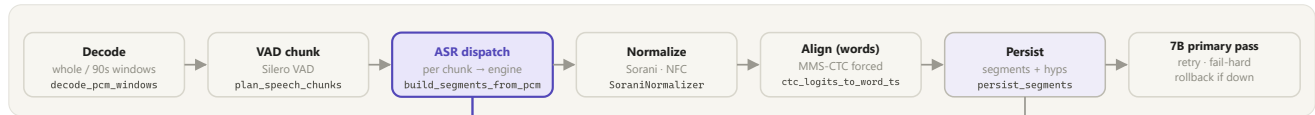
1 · CAPTURE & UI (Svelte 5 frontend)



2 · IPC BRIDGE



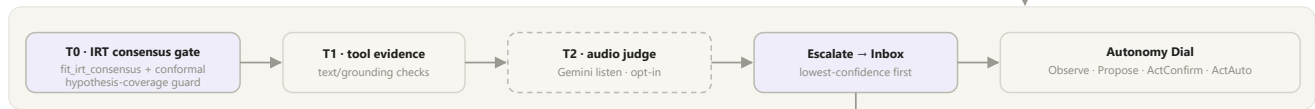
3 · PIPELINE (Rust core — pipeline.rs)



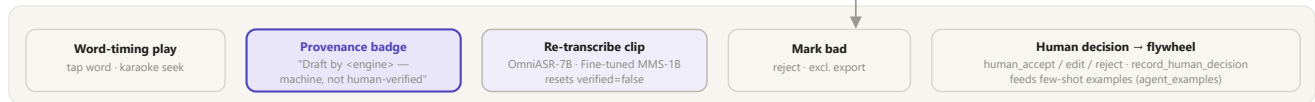
4 · ASR ENGINES (routed by asr_model_size)



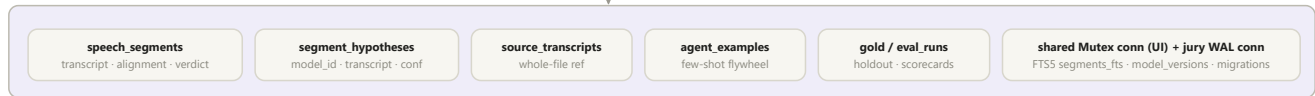
5 · JURY (auto-adjudication, own WAL DB connection — jury/)



6 · HUMAN REVIEW (ReviewMode.svelte — human decision is final, never overwritten)



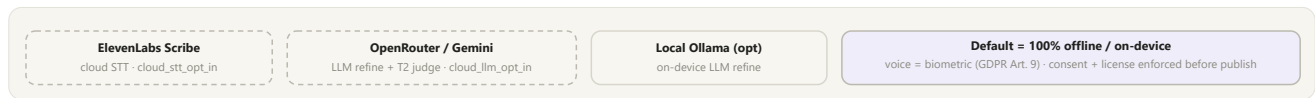
7 · STORAGE (SQLite WAL + FTSS — db.rs · 25 migrations · all transcripts NFC)



8 · VALIDATE & EXPORT (honesty-gated dataset build — export*.rs · eval.rs)



9 · CLOUD (consent-gated, default OFF — effective_llm_mode downgrades cloud→none)



Cortex Speech · architecture as of 2026-06-30 (main @ 4787d81). Solid violet = OmniASR-7B Champion primary path (default). Dashed = cloud, opt-in & off by default.

End-to-end data flow: capture → pipeline → ASR engines → jury → human review → storage → validation & export. Solid violet = the OmniASR-7B Champion primary path (default); dashed = cloud (opt-in, off by default).

1 · What it is

Cortex Speech turns raw long-form audio (podcasts, interviews, audiobooks) into a clean, human-verified, license-aware speech dataset for Central Kurdish — a low-resource language. The workflow is:

import VAD chunk ASR (7B Champion) align words jury adjudicate human review validate export

Design principles

- **Offline-first.** Local ASR + alignment + storage; no network needed.
- **Honesty-first.** Machine output is never presented as human-verified; every transcript carries its engine provenance; metrics come from real runs only.
- **Consent-gated cloud.** Voice is treated as biometric data (GDPR Art. 9); cloud STT/LLM require explicit opt-in.
- **Human-in-the-loop.** A jury auto-adjudicates the easy cases and escalates the rest to a fast review UI; the human decision is final.

Tech stack

- **Shell:** Tauri v2 (Rust core + WebView2 UI).
- **Frontend:** Svelte 5 (runes), ~102 IPC commands, EN + CKB (RTL) i18n.
- **Backend:** Rust — pipeline, ASR, jury, export, storage.
- **ASR runtimes:** sherpa-onnx (CTC), `ort` ONNX (MMS/aligner), Silero VAD, fairseq2+PEFT (7B via WSL GPU).
- **Storage:** SQLite (WAL) + FTS5 full-text search; transcripts stored NFC.

2 · End-to-end pipeline (Rust core)

Import is dispatched from the IPC commands `import_audio_file` / `import_directory` (commands.rs), which spawn a background worker so the UI never blocks. The `ProcessingPipeline` (pipeline.rs) then runs each file through:

Stage	What happens	Key functions / files
Decode	Audio → 16 kHz mono PCM. Short files decode whole; long files stream in 90 s windows to bound memory.	<code>decode_to_pcm_with_timeout</code> , <code>decode_pcm_windows</code> , <code>should_stream_decode</code> (audio.rs, chunking.rs)
VAD chunk	Silero VAD finds speech regions; regions are merged/split/absorbed to honor min/max segment length (no cutting mid-word, no spanning long silences).	<code>voice_activity_detection</code> , <code>plan_speech_chunks</code> (chunking.rs)
ASR dispatch	Each chunk is transcribed by the active engine (default: OmniASR-7B Champion). Confidence + raw text returned per chunk.	<code>build_segments_from_pcm</code> , <code>transcribe</code> (pipeline.rs)
Normalize	Sorani normalization (numerals, hamza/y-k variants) → <code>normalized_transcript</code> ; all text stored Unicode NFC.	<code>SoraniNormalizer</code> (normalizer.rs)
Align (words)	Forced alignment gives per-word start/end timestamps for "tap-a-word" review. Uses the fine-tuned MMS-CTC; falls back to the bundled aligner / energy heuristic.	<code>align_via_finetuned_mms</code> , <code>ctc_logits_to_word_timestamps</code> (aligner.rs, wav2vec2_asr.rs)
Persist	Segments + multi-model hypotheses written transactionally to SQLite; FTS5 index auto-syncs.	<code>persist_segments</code> , <code>insert_segments_batch</code> (db.rs)
7B primary pass	When the 7B is the engine, each segment is (re-)transcribed via the warm WSL server with retry. Fail-hard: if the server is unreachable the import is cancelled and its segments rolled back — no placeholder library.	<code>run_primary_wsl_pass_for_import</code> (pipeline.rs)

After persistence the import callback triggers the jury on a **separate WAL DB connection** so heavy adjudication never starves the UI's `get_segments`.

3 • ASR engines

Five engines are wired; routing is by `settings.asr_model_size` (default **WSL7B**) plus `use_finetuned_asr` and consent flags.

Engine	model_id	Runtime	Role
OmniASR-7B Champion ★	<code>omniasr-wsl-7b</code>	base omniASR-LLM-7B-v2 + LoRA adapter, fairseq2+PEFT, ~31 GB, WSL GPU server on <code>127.0.0.1:8799</code>	Primary / default. The owner's fine-tuned 7B; produces every import transcript.
OmniASR-CTC 300M	<code>omniasr-ctc-300m</code>	sherpa-onnx (ONNX), ~50 MB	Bundled local fallback / offline default-downloadable model.
OmniASR-CTC 1B	<code>omniasr-ctc-1b</code>	sherpa-onnx (ONNX), ~500 MB	Larger base CTC (opt-in).
Fine-tuned MMS-1B	<code>finetuned-mms-ckb</code>	Wav2Vec2-CTC via <code>ort</code> , ~970 MB, ~18.6% CER	Used for word alignment, and as an opt-in per-clip re-transcribe engine.
ElevenLabs Scribe	<code>scribe-v1</code>	Cloud REST (opt-in)	Cloud STT for a second opinion / jury vote — only with <code>cloud_stt_opt_in</code> .

The "which model" answer. There is no "base vs fine-tuned 7B" ambiguity: the OmniASR-7B Champion is the fine-tuned 7B (base + LoRA). The separate fine-tuned **MMS-1B** is a different, smaller model used mainly for word-timing alignment. Out of the box, transcripts are 100% the 7B Champion.

4 • Jury — automatic adjudication

The jury is a multi-tier confidence router that clears the easy cases automatically and escalates the rest to a human, running on its own DB connection.

- **T0 — IRT consensus gate:** the per-segment hypotheses (7B + the two CTC models) are aligned into a confusion network and voted by an Item-Response-Theory model; a calibrated **conformal threshold** + a hypothesis-coverage guard decide auto-accept vs escalate. `fit_irt_consensus`, `quality/irt.rs`, `jury/mod.rs`.
- **T1 — tool evidence:** text-level grounding / consistency checks.
- **T2 — audio judge (opt-in):** a cloud model (Gemini) *listens* to the clip and adjudicates; gated by `jury_cloud_opt_in`, `jury/t2_listener.rs`.
- **Escalation:** unresolved segments enter the **Review Inbox** ordered lowest-confidence-first; the **Autonomy Dial** (Observe → Propose → ActConfirm → ActAuto) sets how much the agent may do unattended.

For this data the three Sorani ASR models agree only ~40–65% (the two CTC models are architecturally related, so their agreement is correlated and the gate vetoes it) — so the jury mostly escalates, which is why fast human review matters.

5 • Human review

ReviewMode.svelte is a distraction-free, one-clip-at-a-time loop. The human decision is authoritative and is never overwritten by later jury runs.

Review actions

- **Word-timing playback** — tap a word to hear it; karaoke highlight; playback bounded to the clip span.
- **Model-provenance badge** — "Draft by <engine> — machine draft, not human-verified", drawn from recorded hypotheses (never inferred).
- **Re-transcribe this clip** — OmniASR-7B (server) or Fine-tuned MMS-1B (CPU); resets `verified=false`.
- **Mark bad** — records a human *reject*: excluded from export, kept and reversible.

Learning flywheel

- Decisions become `human_accept` / `human_edit` / `human_reject` verdicts.
- Confirmed corrections feed `agent_examples` (few-shot retrieval) so the agent improves on the owner's style.
- A LOOP-0 correction memory can auto-apply confirmed fixes to future transcripts (opt-in).

6 · Storage

SQLite in WAL mode with FTS5 search and 25 versioned migrations. A single shared mutex-guarded connection serves UI commands; the jury uses its own WAL connection to avoid blocking reads during long operations.

Table	Holds
<code>speech_segments</code>	The core row: <code>raw_transcript</code> , <code>normalized_transcript</code> , <code>annotated_transcript</code> , <code>alignment_json</code> (word timings + source window), <code>verified</code> , <code>verdict</code> , <code>human_decision</code> , <code>confidence</code> , audio QC fields, <code>model_version_id</code> , gold flag.
<code>segment_hypotheses</code>	Per-model drafts: (<code>segment_id</code> , <code>model_id</code> , <code>transcript</code> , <code>confidence</code> , <code>model_version_id</code>) — the source of the provenance badge + the jury vote.
<code>source_transcripts</code>	Whole-file reference transcripts (when configured).
<code>agent_examples</code> / <code>gold_segments</code> / <code>eval_runs</code>	Few-shot flywheel, held-out gold sets, evaluation scorecards.
<code>model_versions</code> / <code>schema_migrations</code> / <code>segments_fts</code>	Model registry, migration ledger, full-text index.

7 · Validation & export

Export builds an honesty-gated dataset. Three guardrails apply:

- **Quality gates:** WER/CER thresholds (`enforce_quality_gates`) and a per-segment **training grade** — gold
silver review reject .
- **Leakage guards:** held-out gold clips are excluded by content-hash + path (fail-closed even if the file is missing); HuggingFace splits are **speaker-disjoint** via a union-find guard.
- **Privacy:** curator/local paths are redacted from published datasets.

Formats: **JSON / JSONL**, **CSV** (formula-injection-safe), **Parquet**, **HuggingFace datasets**, **WAV/FLAC** clips with metadata, and **DPO preference pairs** for fine-tuning.

8 · Privacy & consent

The app is 100% offline by default. Cloud is reached only behind explicit, acknowledged opt-in:

- `cloud_stt_opt_in` — ElevenLabs Scribe STT.
- `cloud_llm_opt_in` — OpenRouter / Gemini LLM refinement.
- `jury_cloud_opt_in` — the Gemini T2 audio judge.

`settings.effective_llm_mode()` downgrades cloud → none whenever the relevant flag is off, and the pipeline re-checks consent before building any refiner. API keys are never persisted in tracked files; voice is handled as biometric data with consent + license enforced before any publish/train step.

9 · State as of this build

Recent work landed on `main` (verified by cargo fmt/clippy + 823 Rust tests, typecheck/lint + 129 vitest, 20 Python policy gates, and repo-hygiene):

Commit	Change
<code>48bde15</code>	Integrated a 68-commit hardening line into the branch (30 conflicts reconciled, combining both sides' intent).
<code>1a9ae00</code>	OmniASR-7B Champion forced as the default + fail-hard import (cancel + rollback when the 7B server is down); sanitized client repo-tracked.
<code>4787d81</code>	Review feature: per-clip re-transcribe (7B / MMS-1B), model-provenance badge, mark-bad — verified live on the real app.